INDS-4006 / INDS-4007

Industrial Design Major Design Project (Thesis) 25/26

# Audio AR Platform

Final Report

Design is the potential of possibilities
through adaptation, like living organisms.

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September 8, 2025 - April 20, 2026

# Contents

# 01 Design Brief

Project Purpose

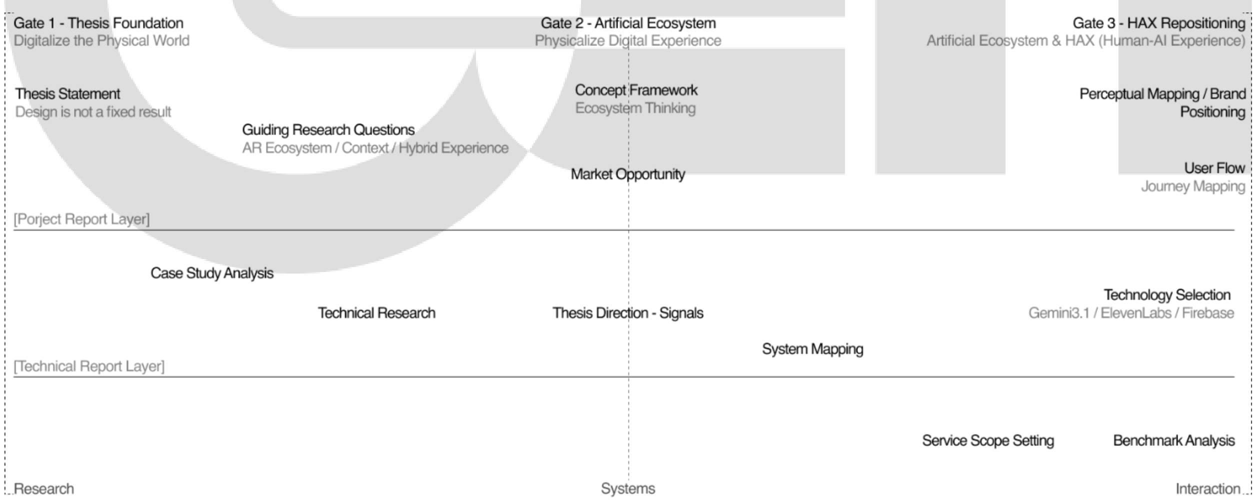
OnGen Radio is an Audio Augmented Reality (AAR) platform that reimagines radio as an intelligent, ambient, context-aware medium for the era of AR glasses and spatial computing. The project repositions radio, long considered an outdated broadcast medium, as a living infrastructure for human–AI experience, where generative AI curates audio content in real time based on who the user is, where they are, and what the moment asks for.

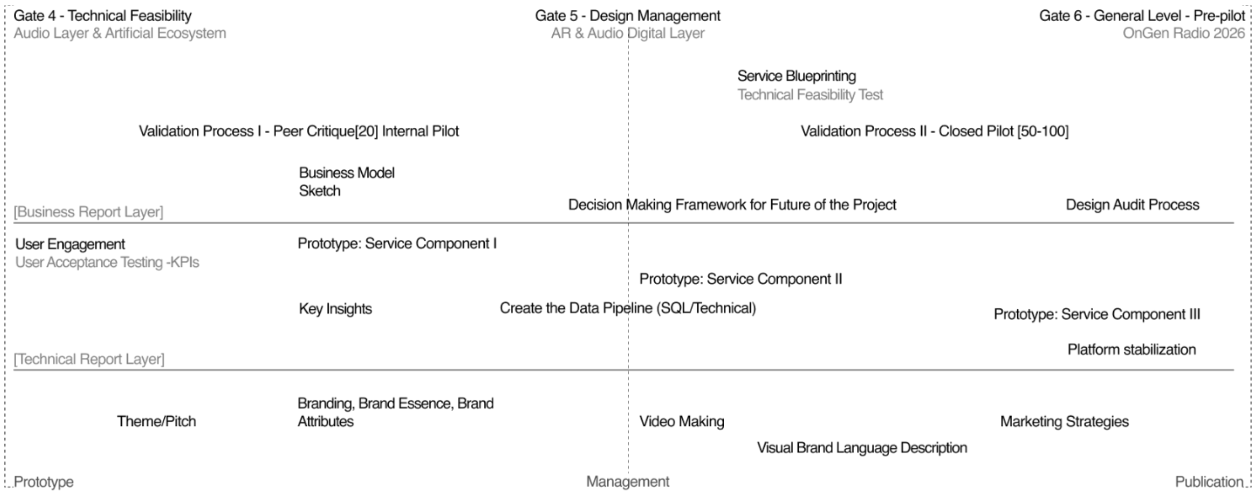
Rather than building another screen-based application, OnGen proposes an audio-first layer that leaves hands and vision free, resists the attention economics of screens, and integrates with the physical environment as a subtle, background intelligence.

## The Problem

Current AI interactions are largely confined to static, reactive conversation: the user asks, the system responds. Meanwhile, algorithmic media feeds demand constant searching, filtering, and manual curation. There is an abundance of information but a scarcity of context. Users switch between a dozen apps to organize their day, and what matters most gets buried beneath what performs best for the algorithm.

At the same time, visual AR has consistently disappointed: it is clunky, battery-hungry, attention-stealing, and slow to achieve mainstream adoption. The structural opportunity sits one layer beneath the eyes, in the audio channel, which is already worn, already intimate, and structurally unoccupied as an AR interface.






Two-phase, six-gate project timeline from thesis foundation to pre-pilot deployment.

## Research Questions

Three guiding research questions shaped the inquiry across strong signals, emerging trends, and weak signals in 2025.

RQ1 How can design improve the understanding of user experience patterns in AR-driven ecosystems?

RQ2 How can we provide personalized services based on environmental context and user state?

RQ3 How can hybrid experience be made intuitive, well-synchronized, and supportive of user communication?

## Thesis Statement

Design is not a fixed result. Design is the potential of possibilities through adaptation, like living organisms — endlessly morphing, with or without human intervention.

## Objectives

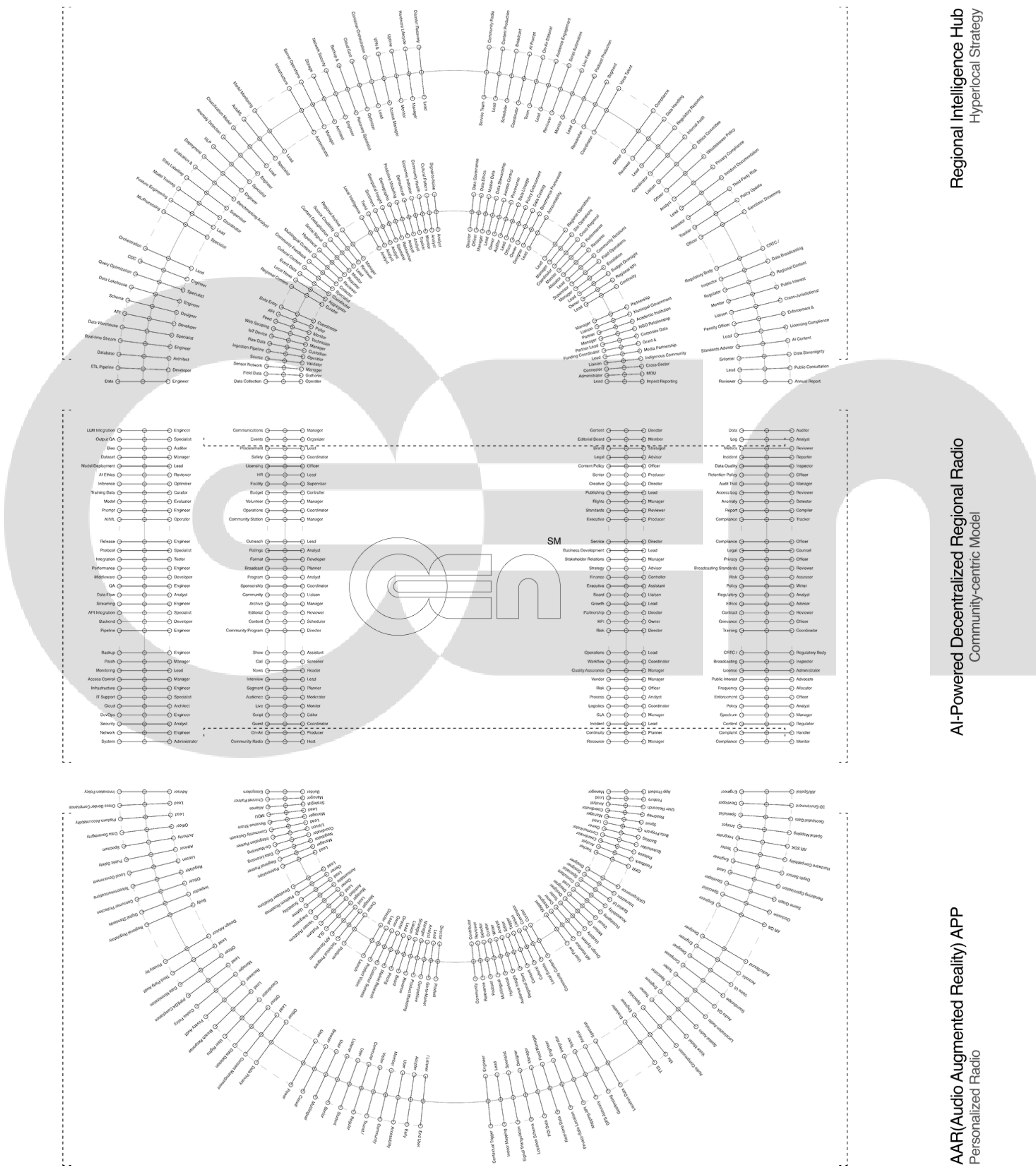
- Reposition the Human–AI Experience (HAX) from static conversation to proactive, dynamic, real-time generative relations.
- Establish audio as a legitimate AR interface and deliver location-aware experiences through existing smartphones before AR hardware reaches mass adoption.
- Build a three-component service system — a Regional Intelligence Hub, an AI-powered community radio, and a personal AAR app — that form a closed feedback loop between physical context and digital broadcast.
- Validate the system through an internal peer critique and a public closed pilot at GradEx 111 in 2026.

## Expected Outcome

A working web-based prototype, a validated service architecture, a brand identity rooted in retro-futurist broadcast heritage, and a phased market entry roadmap positioning OnGen as foundational infrastructure for the Audio-first AR category before AR glasses become mainstream.

## 02 Stakeholders

OnGen's stakeholder system is organized through a RACI ownership framework (Responsible, Accountable, Consulted, Informed) applied independently to each of the three service components. This structure keeps accountability explicit as data moves from physical capture to broadcast to personalized delivery.




High-level map of primary and secondary stakeholders across the Regional Intelligence Hub, Community Radio, and AAR App.

## Primary Stakeholders

Primary stakeholders are directly involved in daily operation and have responsible or accountable roles. They are grouped into three service layers.

| Service Component | Operations & Data | Executive | Technical |
|---|---|---|---|
| Regional Intelligence Hub<br>Hyperlocal Strategy | Data Collection Operator<br>Regional Content Curator<br>Local Intelligence Analyst | Data Governance Director<br>Regional Operations Lead<br>Partnership Manager | Data Engineer<br>ML / Processing Specialist<br>Infrastructure Administrator |
| AI-Powered Decentralized Regional Radio<br>Community-centric Model | Community Radio Host<br>Community Program Director<br>Community Station Manager | Content Director<br>Service Director<br>Operations Lead | System Administrator<br>Pipeline Engineer<br>AI / ML Operator |
| AAR — Audio Augmented Reality App<br>Personalized Radio | App Product Manager<br>UX / Experience Designer<br>Community Content Curator | Product Director<br>Platform Owner (Community)<br>Partnerships Lead | AR / Spatial Engineer<br>Audio / Sound Designer<br>Location Data Specialist |

## Secondary Stakeholders

Secondary stakeholders are informed or consulted parties — end users, regulators, and oversight bodies that receive, audit, or constrain the service.

| Regional Intelligence Hub | Community Radio | AAR App |
|---|---|---|
| Community Radio Service Team<br>Compliance Officer<br>CRTC / Regulatory Body | Data Auditor<br>Compliance Officer<br>CRTC / Regulatory Body | End User / Listener<br>Data Privacy Officer<br>Regional Regulatory Body |

## Value Mapping — The Artificial Ecosystem

Beyond direct stakeholders, OnGen operates within a five-layer value network. Each layer represents a category of external actors that the ecosystem either depends on or serves.

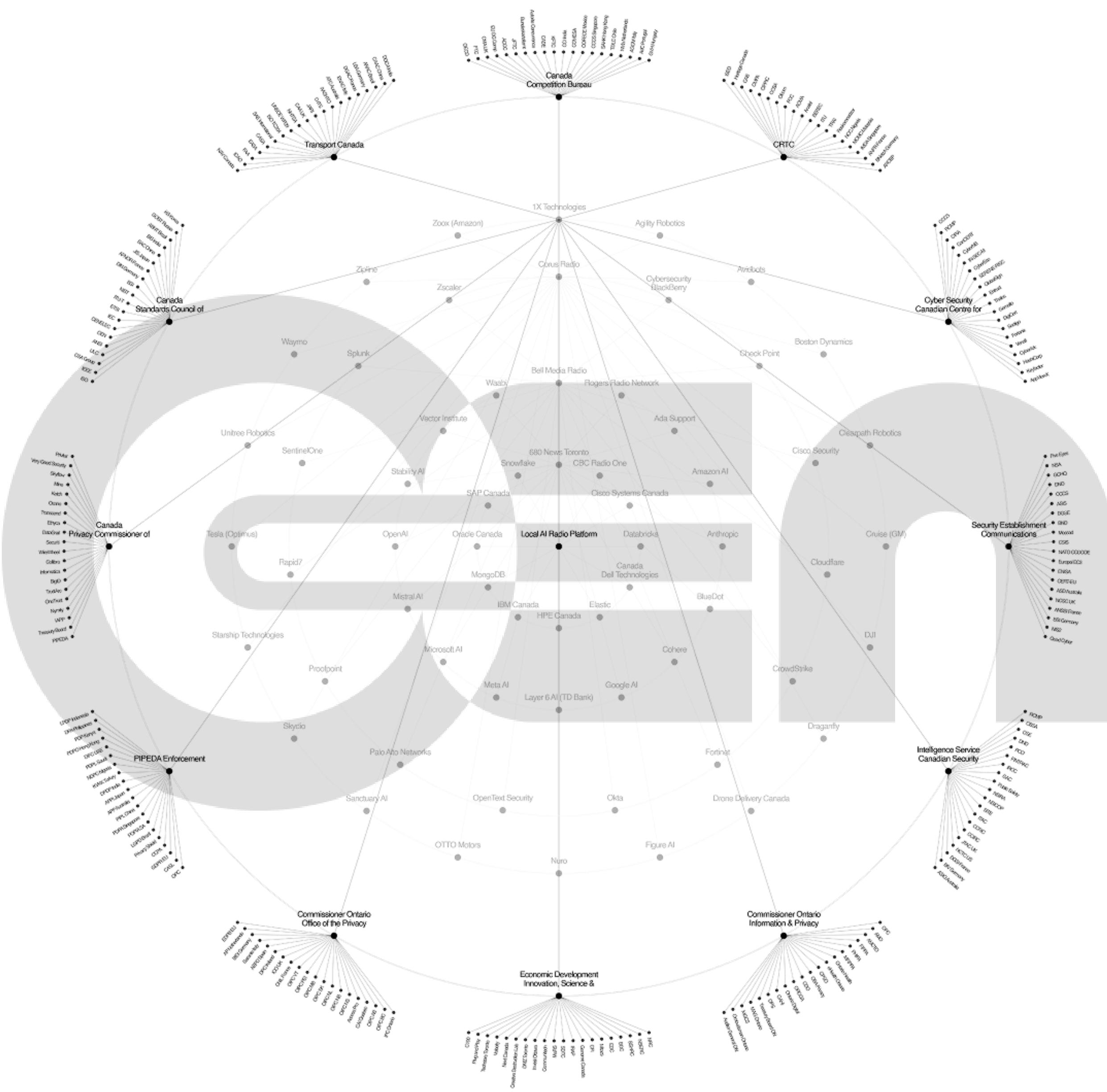
Layer I Data Infrastructure — cloud and database providers (AWS Canada, Oracle Canada, SAP, Cisco, Snowflake, Databricks).

Layer II Data Operators — AI — OpenAI, Anthropic, Google AI, Microsoft AI, Meta AI, Cohere, Mistral, Vector Institute.

Layer III Data Security — CrowdStrike, Palo Alto, Fortinet, Okta, Zscaler, Cloudflare, SentinelOne.

Layer IV Local Entities — civic institutions, community venues, retail, logistics, and cultural partners.

Layer V  Legal Conditions — CRTC, OPC, Privacy Commissioner of Canada, Canadian Centre for Cyber Security, PIPEDA enforcement.




The five concentric layers that situate OnGen within its data, intelligence, security, civic, and regulatory environments.

## 03 User Groups

OnGen's three user archetypes were derived from the AAR Beta Survey and refined through iterative persona work. Each persona represents a distinct relationship with audio, context, and technology, and together they cover the full adoption curve from ambient listener to early AR adopter.

### Persona I — Maya Chen

The Wanderer · 27 · Toronto (Queen West) · Media Studies, University of Toronto · 38.72 Hz

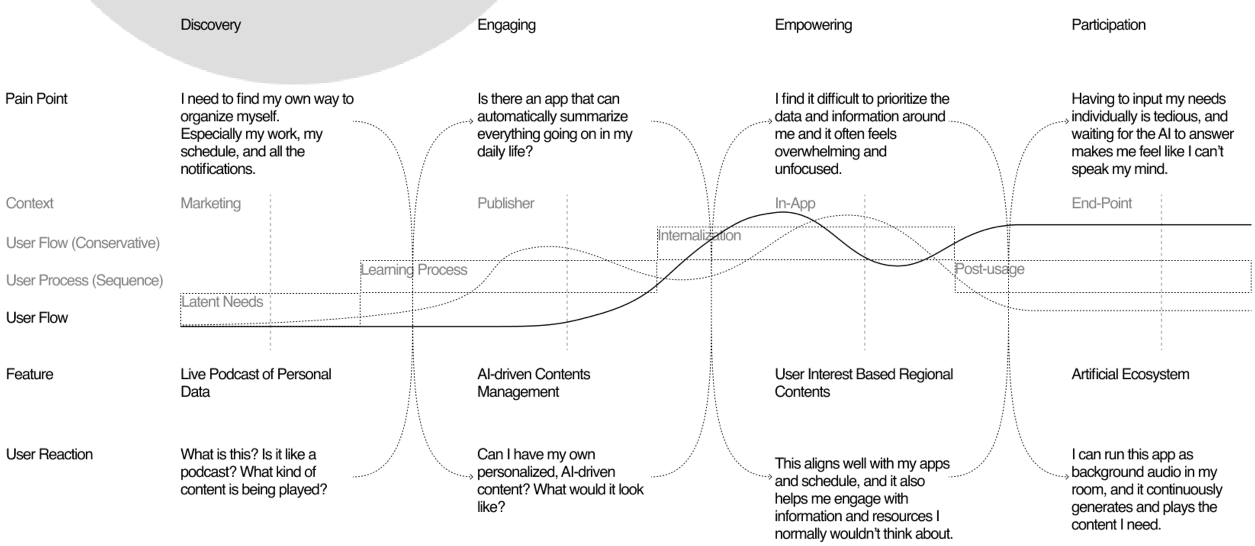
Maya walks through the city with earbuds in, treating streets like a soundtrack to her day. She resists the algorithmic feed but still wants to discover what is happening nearby — exhibitions, pop-ups, community events. For her, the city is a living text she would rather listen to than scroll through.

Motivations

- Discover local culture without active searching.
- Replace doomscrolling with ambient information.
- Feel rooted in the neighbourhood she moves through.
- Keep her walks screen-free but informed.

Pain Points

- Information overload across apps.
- Missing local events until after they happen.
- Navigation apps feel cold and transactional.
- Discovery feels like work, not serendipity.



## Persona II — Jordan Campbell

The Connector · 31 · Toronto (Kensington Market) · Business Admin, Toronto Metropolitan U. · 107.06 Hz

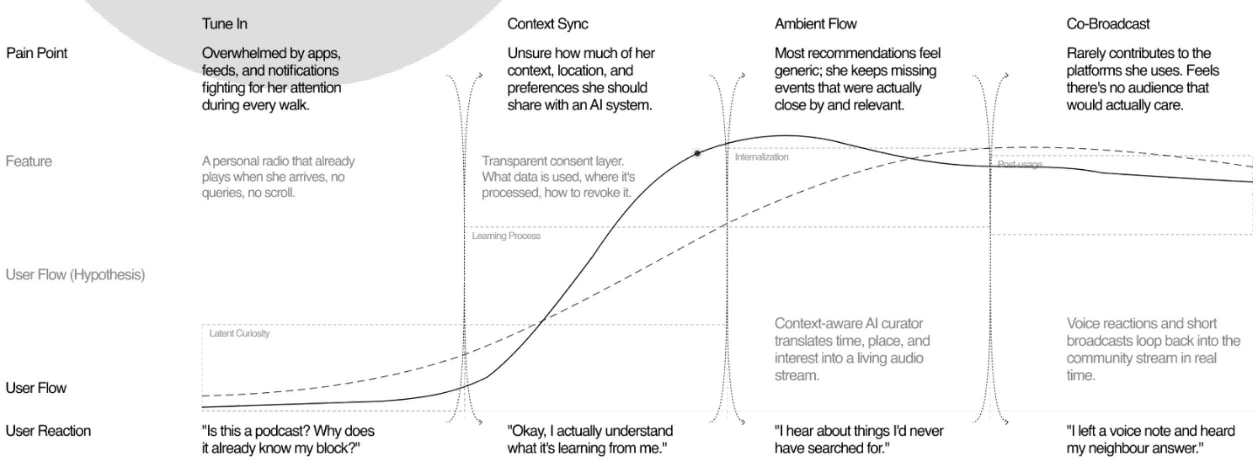
Jordan runs a coworking space in Kensington and organizes open-mic nights, pop-ups, and monthly socials. His work is making strangers feel like regulars. Social platforms dilute the local culture he is trying to build; he wants a channel that belongs to the neighbourhood, not to an algorithm three thousand kilometres away.

### Motivations

- Amplify the voices of the people he works alongside.
- Coordinate events with minimal friction.
- Preserve the identity of Kensington culture.
- Give members a reason to show up in person.

### Pain Points

- Flyers and posters buried by city noise.
- Instagram reach keeps shrinking for local posts.
- Members miss announcements across channels.
- No shared broadcast layer exists for the block.



## Persona III — Alex Rivera

The Pioneer · 35 · Toronto (Distillery District) · HCI, Carnegie Mellon University · 86.51 Hz

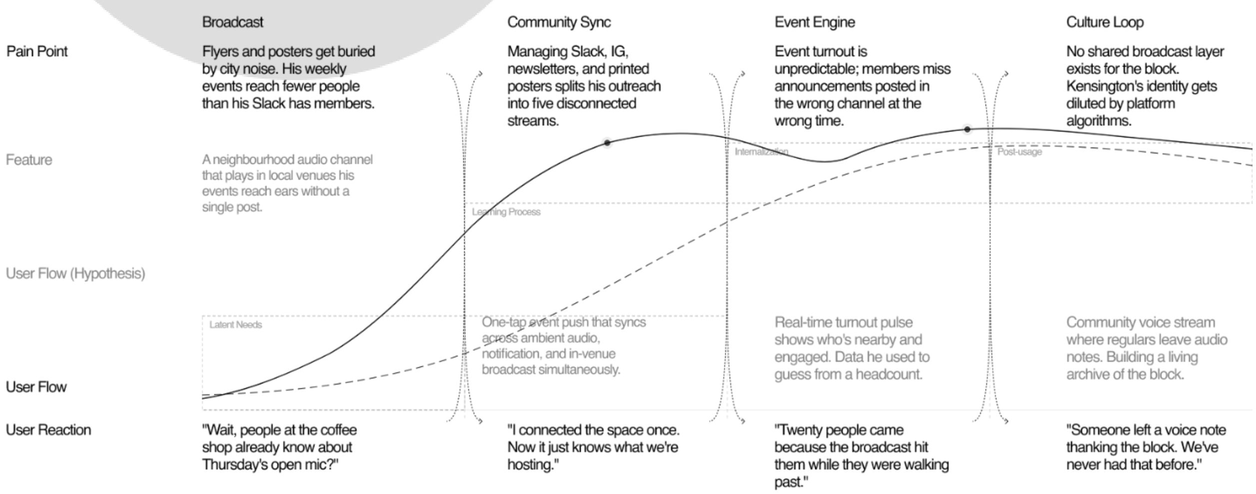
Alex has worn smart glasses daily since the first Ray-Bans launched. She designs products for a living and tests every new AR / AI device that ships. Visual AR has disappointed her — clunky, battery-hungry, attention-stealing. She is convinced the real breakthrough is audio-first AR, where the interface fades into the environment.

### Motivations

- Interact with AI hands-free, eyes-free.
- Stay ahead of emerging interaction paradigms.
- Reduce screen time without losing capability.
- Test audio-AR as a serious productivity layer.

### Pain Points

- Visual AR disrupts focus and drains battery.
- Voice assistants only respond when summoned.
- No ambient AI layer between apps and reality.
- Context always has to be manually entered.



# 04 Interviews, Testing, and Results

AAR Beta Survey – Your Community, Your Sound

Primary research took the form of a qualitative beta survey conducted from March 11 to March 25, 2025, with consented participants aged 13 and over. The survey measured current audio behaviour, openness to AI curation, appetite for local content, and concerns around privacy and sensory overstimulation.

## Core Findings

| | |
|---|---|
| 91% | of respondents listen to audio content daily — audio-first habits are already established. |
| 91% | are open to receiving local audio; music streaming and podcasts lead the way as distribution channels. |
| 59% | want local information and community-relevant content surfaced in their audio feed. |
| 55% | regularly consume podcasts, providing an adjacent behavioural bridge to AI-curated spoken audio. |
| 3.36 / 5 | average sentiment toward AI curation — cautiously positive, and moving upward among frequent AI users. |
| 3.41 / 5 | average satisfaction with current community-information access, indicating a clear gap to close. |
| 3.00 / 5 | comfort with AI data handling — the lowest score, and the most important signal to design against. |

## Participant Feedback

### Excitement

> Interest-based curation would be a big help, as I tend to miss a lot of events.

> Keep me in the loop with relevant info while I've got my music on.

### Concern

> To what extent does it pick up on daily activities or private conversations?

> Unwanted audio can be overstimulating and anxiety-inducing.

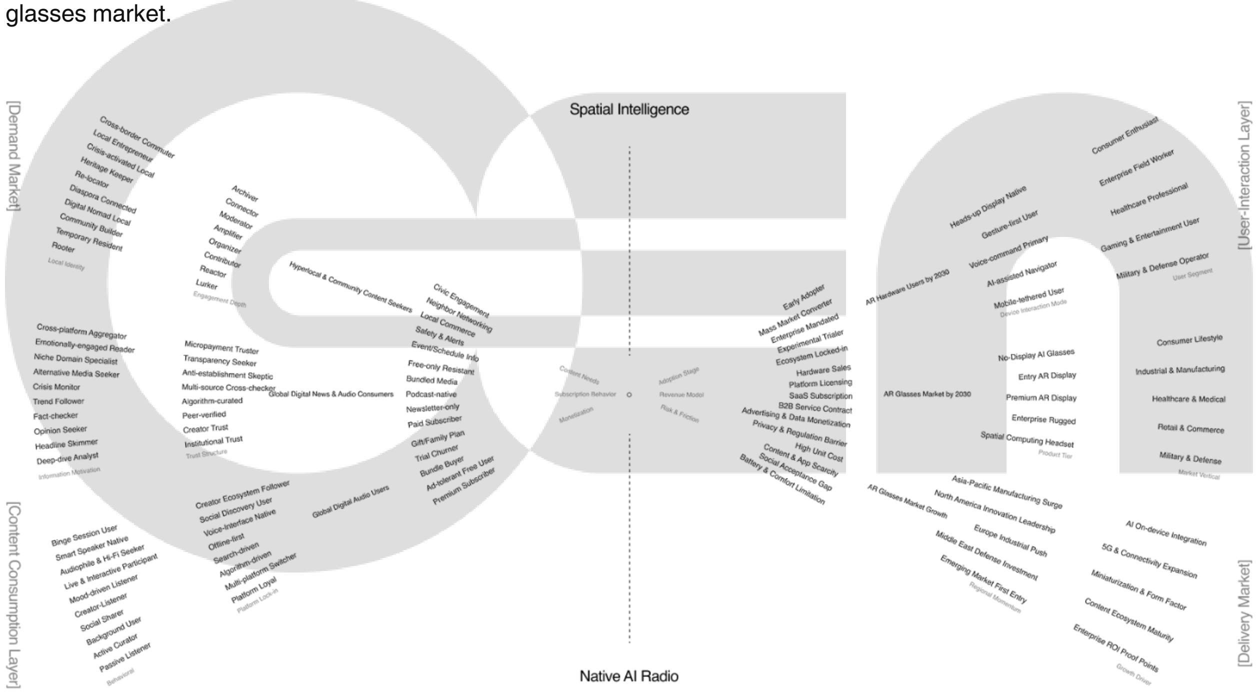
## Key Insight

Participants highly value interaction-based, contextual relevance but express critical concerns around sensory overstimulation and privacy intrusion. Design must earn trust through transparent consent, local processing, and user-controlled intervention — concerns that directly shaped the Privacy-First Localized AI feedback loop described in Section 5.

## Market Validation

Primary research was triangulated against market data. Statistics Canada's 2025 Q2 survey of AI adoption among Canadian businesses shows Text Analytics (35.7%), Data Analytics (26.4%), Virtual Agents / Chatbots (24.8%), Natural Language Processing (23.1%), and Speech Recognition (20.0%) as the leading AI categories — a landscape still dominated by static, chatbot-based interactions, not dynamic live-audio ones.

Three market projections confirm the size of the opportunity: AI in Telecommunications is projected to grow from USD 4.78B (2025) to USD 58B (2032); Cognitive Radio from USD 8B (2025) to USD 58B (2035); and the AI API market from USD 44.4B (2025) to USD 180B (2030). AR hardware users are forecast to reach 160M by 2030 with a USD 10B AR glasses market.



## 05 Consumer Journeys — Now, Next, Wow

The Now, Next, Wow framework maps the evolution of the audio-context experience across three horizons: the current algorithmic present, the improved OnGen-launched experience, and the long-term aspirational layer that arrives with AR glasses.

### Now · Present State (2025)

Current User Reality - Algorithm

There is plenty of information, but very little context. Users switch between a dozen apps and algorithms, each demanding manual interaction: constant searching and filtering, with a feed that never ends yet keeps missing what actually matters.

- Interactions with AI are reactive — ask and receive, nothing more.
- Content is pushed by engagement metrics, not proximity or relevance.
- Cognitive load is high; ambient awareness is low.

### Next · Improved Experience (2026)

OnGen Radio Launch - Navigation

Context listens, and the signal becomes a channel. OnGen Radio listens to the user's context and translates it into their own language. Content plays automatically at the moment it is wanted — no apps to open, no queries to type.

- A personal radio already plays when the user arrives in a location.
- A transparent consent layer shows exactly what data is used, where it is processed, and how to revoke it.
- The AI becomes a proactive curator rather than a reactive respondent.

### Wow · Aspirational Transformation (2028+)

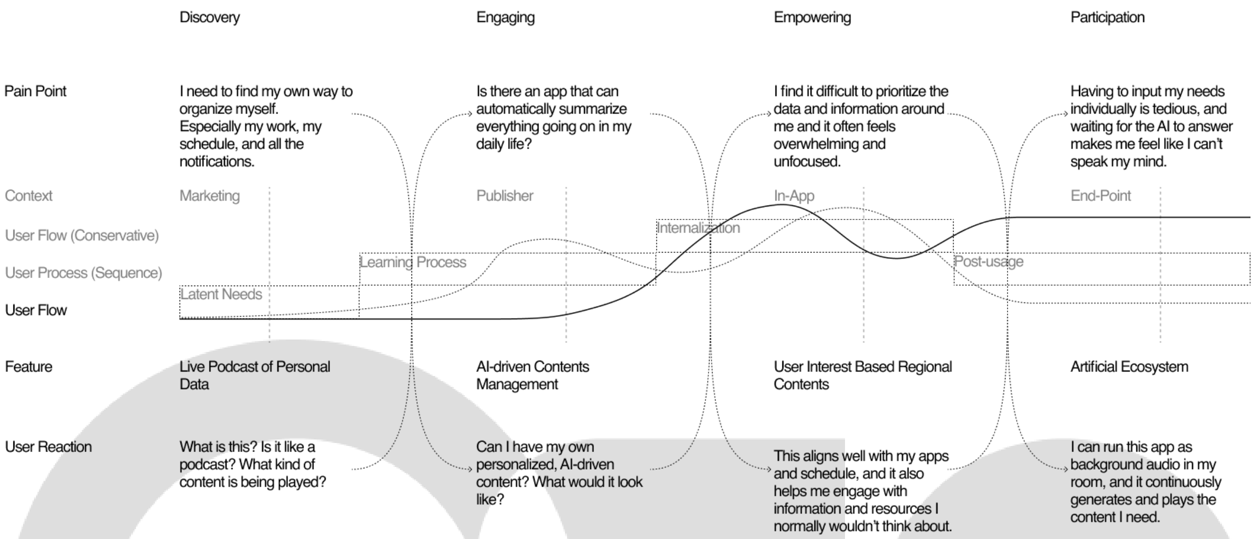
The Era of AR Glasses - Native / Spatial AI

The city speaks to the user as they walk. The radio becomes a space. In the era of AR glasses, OnGen is no longer an app to open but an ambient layer that lives in the places the user moves through. Every street has its frequency. Every community has its voice.

- Audio AR is recognized as a category, with consumer mass established.
- Ambient commerce, hyperlocal advertising, and spatial identity layers emerge on top of the radio infrastructure.
- The boundary between physical and digital dissolves; spatial radio becomes foundational infrastructure, as universal and invisible as connectivity itself.

## User Journey Flow — Hypothesis

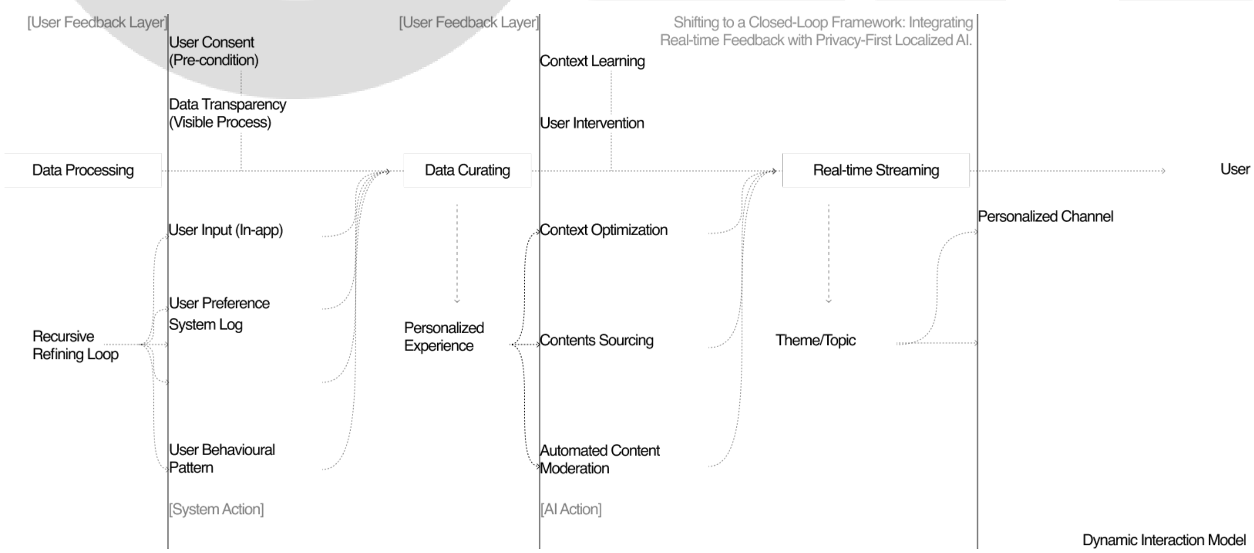
The consolidated hypothesis flow maps the user across four progressive stages — Discovery, Engaging, Empowering, and Participation — tracking pain points, context, features, and anticipated user reactions at each stage. It is the structural spine that connects the Now / Next / Wow horizons above to the scenario storyboards that follow.



Pain Point × Context × Feature × User Reaction across four journey stages.

## HAX Repositioning — Feedback Loop

Across all three horizons, OnGen replaces the static chatbot loop with a closed-loop, privacy-first, localized AI framework. User Consent is a precondition, Data Transparency is a visible process, and the system runs on a recursive refining loop where user input, behavioural pattern, and preference feed continuously into real-time streaming and personalized channel curation.



Data processing, curating, and real-time streaming are mediated by a user feedback layer across consent, transparency, input, preference, system log, and behaviour.

## 06 Experience Scenarios

Three short scenarios demonstrate OnGen in use, one per persona. Each scenario is structured as a linear narrative across four moments: a cold first contact, a deepening sync, a moment of delivered value, and a closing loop back into the community.

### Scenario I — Maya's Walk

Urban Explorer

Maya leaves her apartment for a 45-minute walk through Queen West. Her Ray-Bans and phone are already paired. She taps nothing.

#### Tune In

A personal radio already plays when she arrives outside — no queries, no scroll. Maya reacts: "Is this a podcast? Why does it already know my block?"

#### Context Sync

A transparent consent layer shows what data is being used, where it is being processed, and how to revoke it. Maya reacts: "Okay, I actually understand what it's learning from me."

#### Ambient Flow

A context-aware AI curator translates time, place, and interest into a living audio stream. Maya reacts: "I hear about things I'd never have searched for."

#### Co-Broadcast

Voice reactions and short broadcasts loop back into the community stream in real time. Maya reacts: "I left a voice note and heard my neighbour answer."

Urban Explorer / Graduate Student

Age

27

Location

Toronto, ON
Queen West

Education

Media Studies
University of Toronto

Archetype

The Wanderer

Username

38.72hz

Background

Maya walks through the city with earbuds in, treating streets like a soundtrack to her day. She resists the algorithmic feed but still wants to discover what's happening nearby: exhibitions, pop-ups, community events. For her, the city is a living text she'd rather listen to than scroll through.

Motivation

- Discover local culture without active searching
- Replace doomscrolling with ambient information
- Feel rooted in the neighborhood she moves through
- Keep her walks screen-free but informed

Pain Points

- Information overload across apps
- Missing local events until after they happen
- Navigation apps feel cold and transactional
- Discovery feels like work, not serendipity

Daily Behaviors

- Walks 45+ min daily, always with AirPods
- Listens to podcasts, indie music, ambient sets
- Checks Instagram maps for "what's nearby"
- Saves events she forgets to attend

Technology

## Scenario II — Jordan's Open Mic

Community Connector

Jordan runs a Thursday open-mic at his Kensington coworking space. Turnout used to depend on flyers and Slack.

### Broadcast

A neighbourhood audio channel plays in local venues; his event reaches ears without a single post. Jordan reacts: "Wait, people at the coffee shop already know about Thursday's open mic?"

### Community Sync

A one-tap event push syncs across ambient audio, notification, and in-venue broadcast simultaneously. Jordan reacts: "I connected the space once. Now it just knows what we're hosting."

### Event Engine

A real-time turnout pulse shows who is nearby and engaged — data he used to guess from a headcount. Jordan reacts: "Twenty people came because the broadcast hit them while they were walking past."

### Culture Loop

A community voice stream lets regulars leave audio notes, building a living archive of the block. Jordan reacts: "Someone left a voice note thanking the block. We've never had that before."

Community Member / Coworking Space Manager

Age

31

Location

Toronto, ON
Kensington Market

Education

Business Admin
Toronto
Metropolitan U.

Archetype

The Connector

Username

107.06hz

#### Background

Jordan runs a coworking space in Kensington and organizes open-mic nights, pop-ups, and monthly socials. His work is making strangers feel like regulars. Social platforms dilute the local culture he's trying to build; he wants a channel that belongs to the neighborhood, not to an algorithm 3,000 km away.

#### Motivation

- Amplify voices of the people he works alongside
- Coordinate events with minimal friction
- Preserve the identity of Kensington culture
- Give members a reason to show up in person

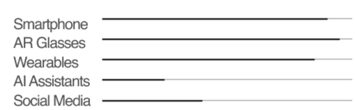
#### Pain Points

- Flyers and posters buried by city noise
- Instagram reach keeps shrinking for local posts
- Members miss announcements across channels
- No shared broadcast layer for the block

#### Daily Behaviors

- Hosts 2–3 community events every month
- Manages Slack, IG, newsletter, and posters
- Chats with every regular who walks in
- Tracks turnout in a spreadsheet by hand

#### Technology



## Scenario III — Alex's Ambient Layer

AAR Early Adopter

Alex wears Meta Ray-Bans every day and has a conference demo in the Distillery District.

### Device Pair

Instant smart-glasses pairing. OnGen detects her Ray-Bans and begins ambient audio without a setup wizard. Alex reacts: "It recognized my Ray-Bans in three seconds. No app, no wizard, just audio."

### Ambient Sync

A zero-input context engine reads location, time, movement, and calendar to compose a relevant audio stream. Alex reacts: "I didn't configure anything and it already knows I'm heading to the Distillery."

### AI Layer

A proactive AI curator surfaces information at the right moment; audio fades in without interrupting focus. Alex reacts: "A gallery opening whispered into my walk. I would've scrolled past it on any screen."

### Interface Fade

A persistent ambient layer spans glasses, phone, and watch as one unified auditory interface. Alex reacts: "This is the ambient AI layer I've been prototyping in Figma. Someone built it."

#### AAR Early Adopter / Senior UX Designer

Age

35

Location

Toronto, ON
Distillery District

Education

HCI
Carnegie Mellon
University

Archetype

The Pioneer

Username

86.51hz

#### Background

Alex has been wearing smart glasses daily since the first Ray-Bans launched. She designs products for a living and tests every new AR/AI device that ships. Visual AR has disappointed her: clunky, battery-hungry, attention-stealing. She is convinced the real breakthrough is audio-first AR, where the interface fades into the environment.

#### Motivation

- Interact with AI hands-free, eyes-free
- Stay ahead of emerging interaction paradigms
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#### Pain Points

- Visual AR disrupts focus and drains battery
- Voice assistants only respond when summoned
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- Context is always manually entered

#### Daily Behaviors

- Walks 45+ min daily, always with AirPods
- Listens to podcasts, indie music, ambient sets
- Checks Instagram maps for "what's nearby"
- Saves events she forgets to attend

#### Technology

Smartphone
AR Glasses
Wearables
AI Assistants
Social Media

# 07 Perceptual Map

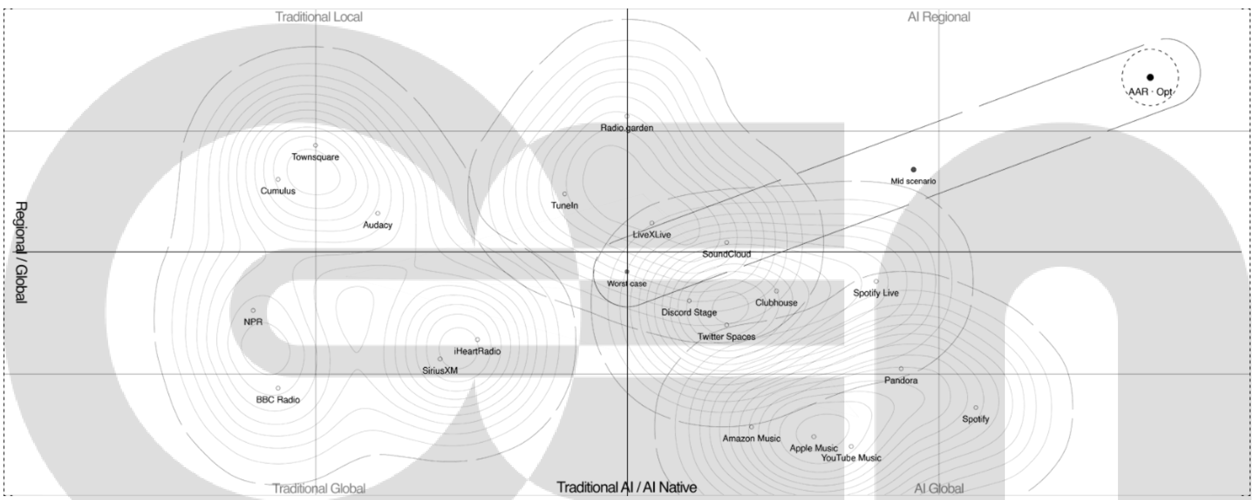
OnGen's competitive position is mapped across three distinct markets. Each map uses a two-axis frame to expose where incumbents cluster and where structural whitespace sits.

## Demand Market — Audio / Radio Platform

Axes Regional ↔ Global · Traditional ↔ AI

Streaming giants have AI but no local context. Regional broadcasters have local presence but no AI capability. Neither can cross into the AI-Regional quadrant without rebuilding from scratch. OnGen occupies this whitespace as an ambient local layer that converts real-time regional signals into audio, not as a better radio app.

Position AI × Regional quadrant — Intelligence Hub operational, multi-region networks live, no credible competitor.

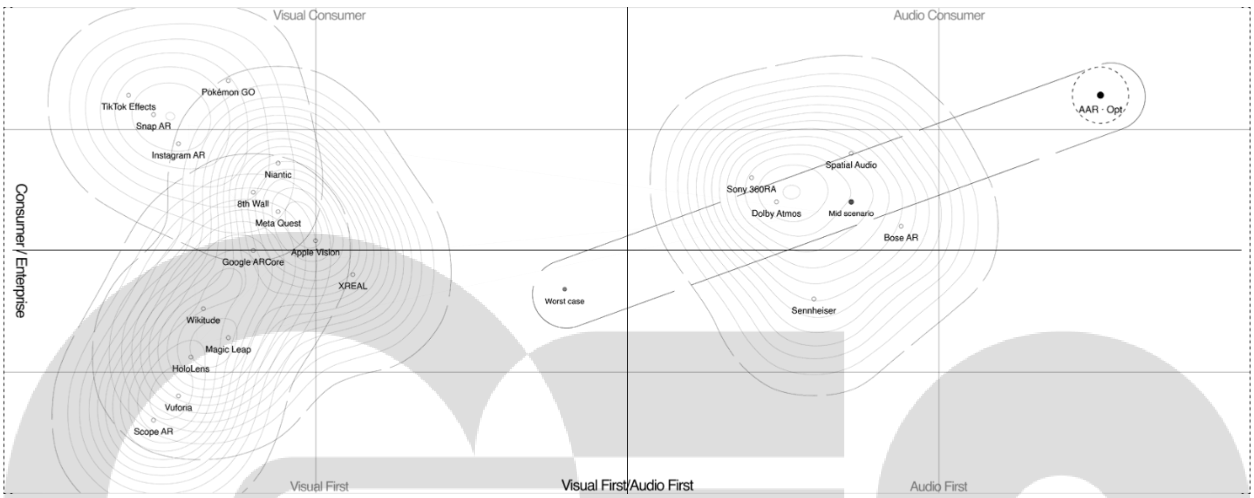


## Delivery Market — AR / XR

Axes Consumer ↔ Enterprise · Visual ↔ Audio

Almost all AR investment targets the visual. Audio as an AR interface is structurally unoccupied. OnGen delivers location-aware audio experiences with only GPS and a smartphone — the only AR model with immediate mass-consumer reach today.

Position Audio-First × Consumer quadrant — recognized as a new category; strong early-adopter signal with education required for mainstream diffusion.

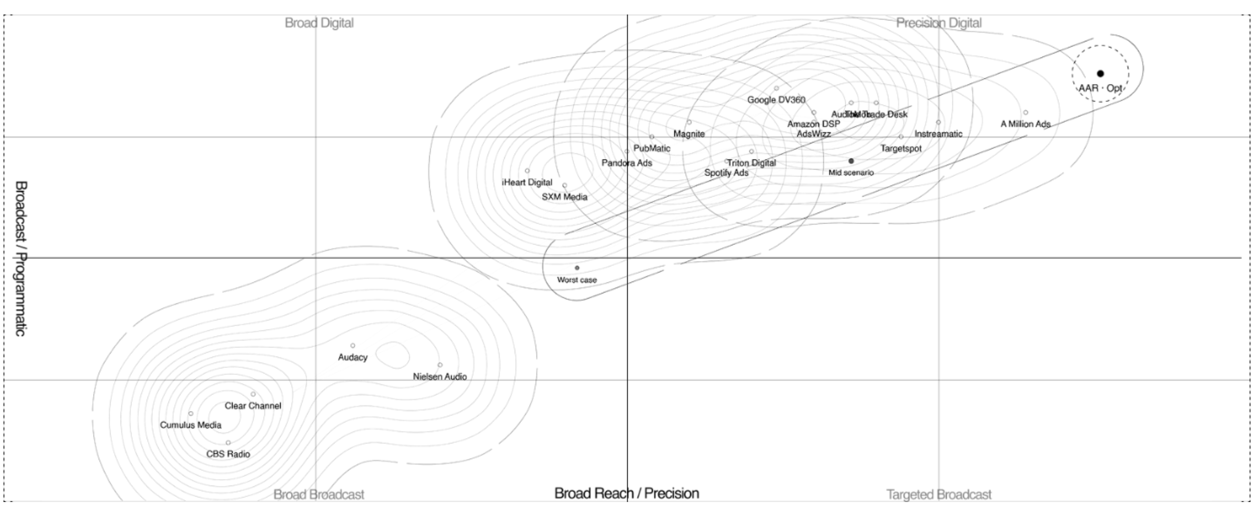



## Revenue Market — Programmatic Audio Ad

Axes Broadcast ↔ Programmatic · Broad ↔ Precision

Traditional broadcast has reach but no targeting. Digital audio has programmatic capability but no local context. OnGen feeds real-time signals — events, weather, foot traffic — into ad insertion, creating a hyperlocal CPM tier that does not currently exist.

Position Targeted Broadcast quadrant — programmatic + location-precision + regional advertiser network.



# 08 Brand Values

Built on the legacy of broadcast, powered by real-time AI generation.

OnGen's identity emerges from five commitments that shape its tone, visuals, and user experience. Each value is expressed not only through design but through how the service listens, speaks, and restrains itself.

## Contextual

Every signal is bound to place, time, and self. Content emerges from where users are, who they are with, and what the moment asks for. OnGen refuses to be content-agnostic.

## Ambient

A layer beneath the eyes, not in front of them. Audio-first, background by default. OnGen leaves hands and vision free, resisting the attention economics of the screen.

## Living

Design that morphs with and without intervention. The system adapts continuously to context and feedback, never settling into a fixed final state.

## Collective

The signal is shared, the voice is many. Every listener is a potential broadcaster. Community stories cycle back into the stream. Participation is structural, not decorative.

## Native

Built for AR, not retrofitted into it. Generative, real-time, spatially aware. OnGen is designed for an audio-first AR era, not as a port from existing screen paradigms.

## Visual Identity Reference

The visual system is rooted in the language of 1970s broadcast radio, retro-futurism, and continuity. The palette is monochromatic by design. Typography carries the system; color stays out of the way.

### Visual Language

- Iconicity drawn from podcast and transmission — the dial, the on-air marker, the On/Off power button.
- Skeuomorphism used deliberately, as a bridge between broadcast memory and AI generation.
- Bauhaus-based layout with black-weight typography as the structural anchor.

### Typography

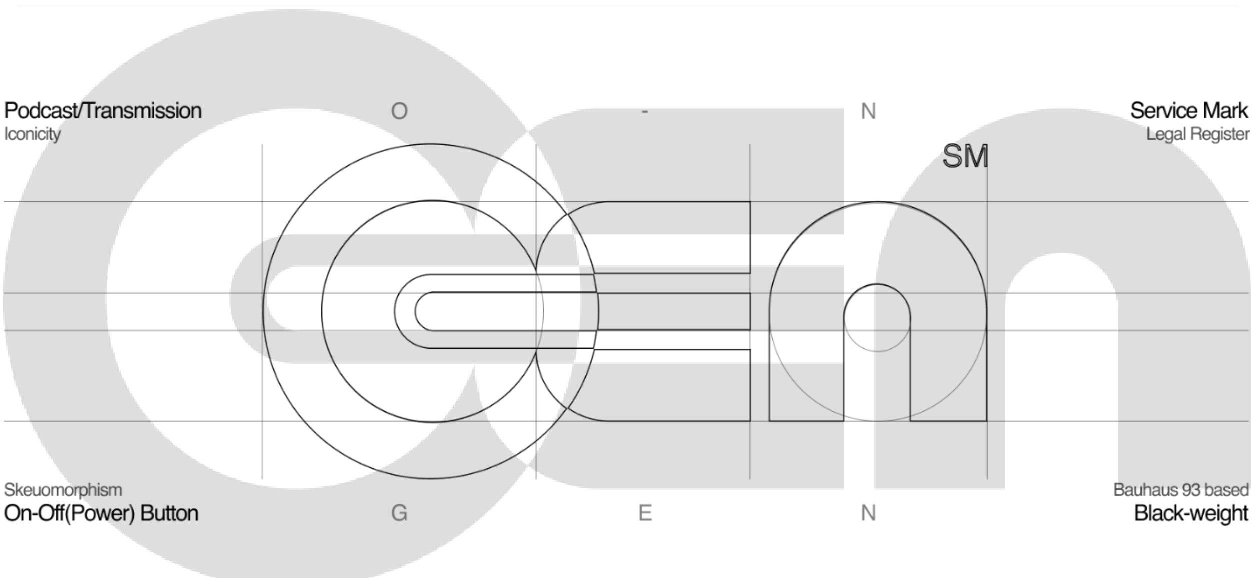
- Primary typeface · Helvetica Regular.
- Display weights · Bold reserved for broadcast cues and hierarchical anchors.
- No decorative color accents in body text; all emphasis through weight and spacing.

### Palette

- Ink · near-black (#111111) · all primary text and line work.
- Muted · neutral grey (#666666) · secondary information and captions.
- Line · light grey (#CCCCCC) · rules, dividers, borders.

Typography · Helvetica Neue

| | |
|---|---|
| Display 01 | Living Frequency |
| Display 02 | Native AI Radio in AR |
| Body | A radio that listens to you, so it knows what to play next. |
| Meta | Regular / 400 · Light / 300 · Letter-spacing –0.05em |




The identity holds one broadcast signal-line as its signature gesture; the type does the heavy lifting.

# 09 Prototype and Experiment

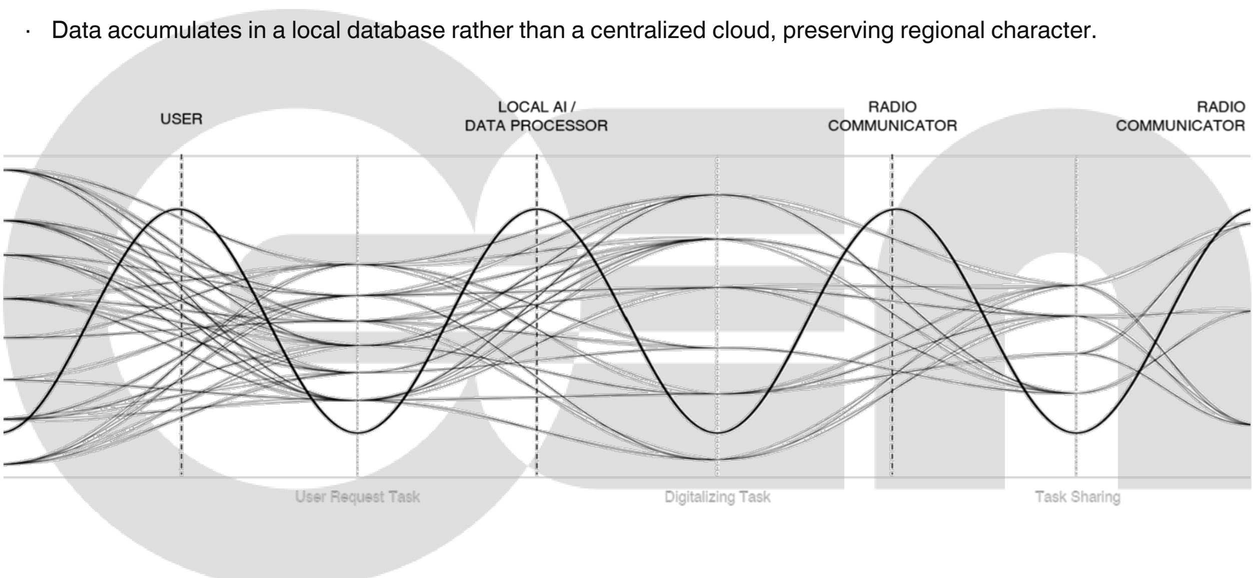
The project operates across three tightly coupled service components, each prototyped as a web application during Phase II and scheduled for a public closed pilot at GradEx 111.

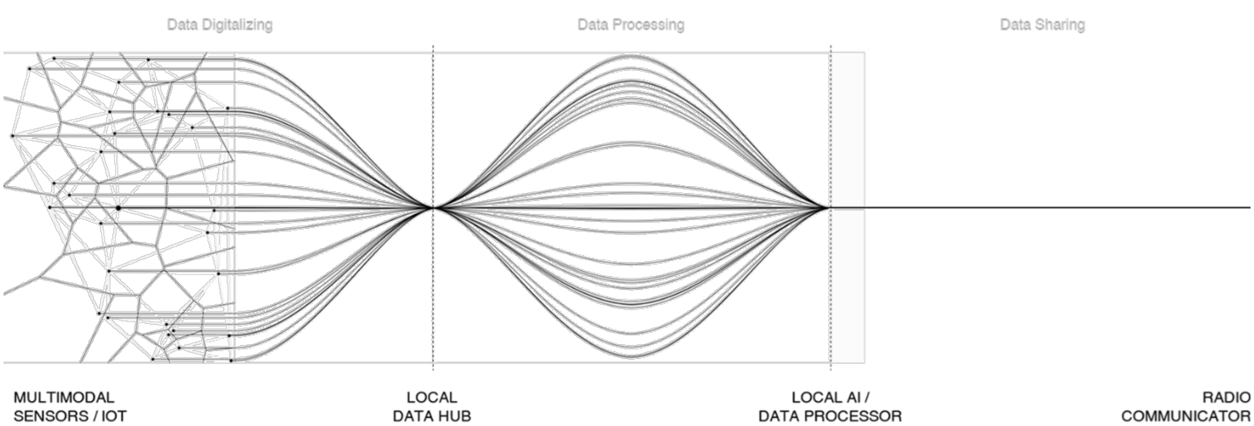
## Component I — Regional Intelligence Hub

### Hyperlocal Strategy

Within an urban environment, diverse local entities access information from across the city, including accessibility and movement data, acquiring and reflecting real-time context about routing and the physical environment. This allows them to maximize the efficiency of AI systems that depend on continuously updated, real-time inputs.

- Physical-based local database, ingesting geographic, civic, and cultural data in real time.
- Compression and parsing algorithms convert digitized physical data into lightweight transmissions.
- Data accumulates in a local database rather than a centralized cloud, preserving regional character.



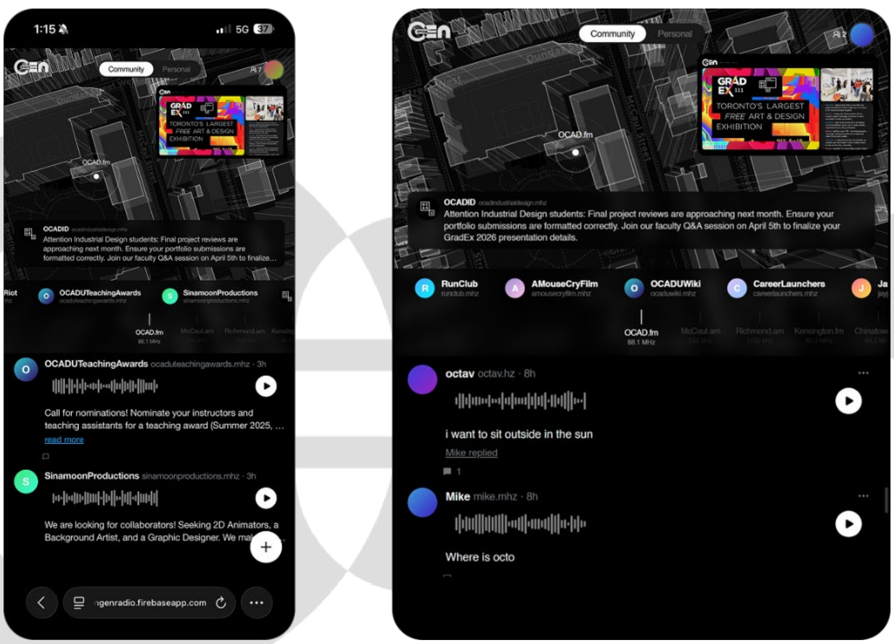



## Component II — AI-Powered Decentralized Regional Radio

### Community-centric Model

OnGen Radio builds on the Hub to power a new kind of community radio where real voices drive the signal and AI handles the architecture behind it. Anyone can broadcast. Every listener is guided. The network grows stronger the more people use it. Not a feed. Not an algorithm. A living frequency.

- Open community space — visitors discover what communities exist and what they offer.
- Channels and feedback loop — community stories → streaming → reactions → streaming.
- Local information surfacing — events, issues, and cultural elements that digital platforms rarely capture.
- User reactions cycle back into the live feed, driving digital participation in the web-app pilot.



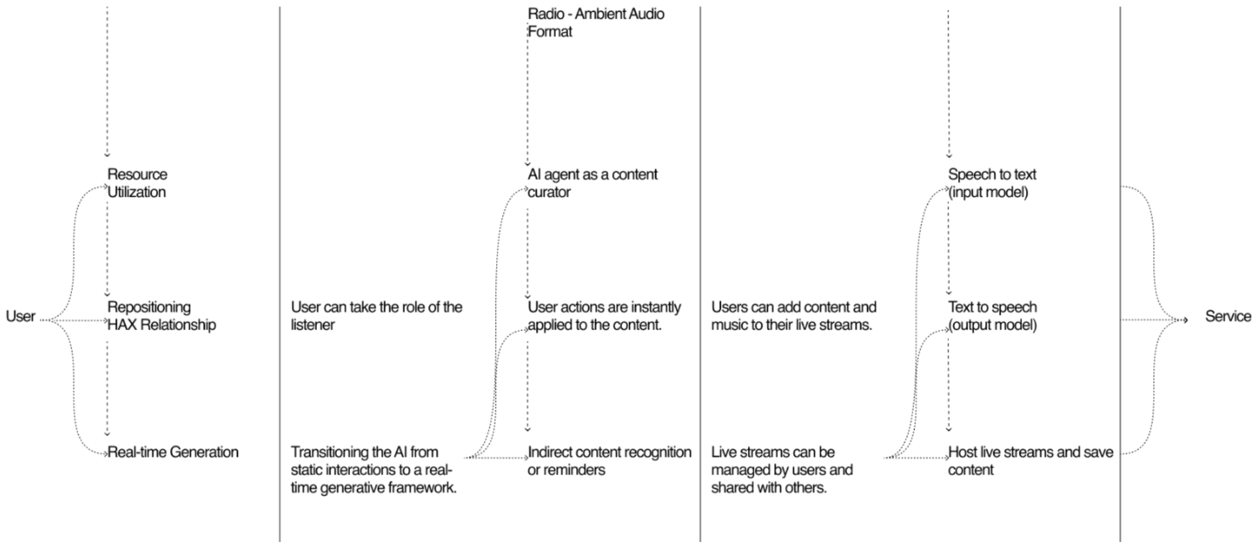

Prototype at ongenradio.web.app — the first public-facing build of the Community Radio component.

## Component III — Personal Radio / AAR

### Personalized Radio – Audio Augmented Reality

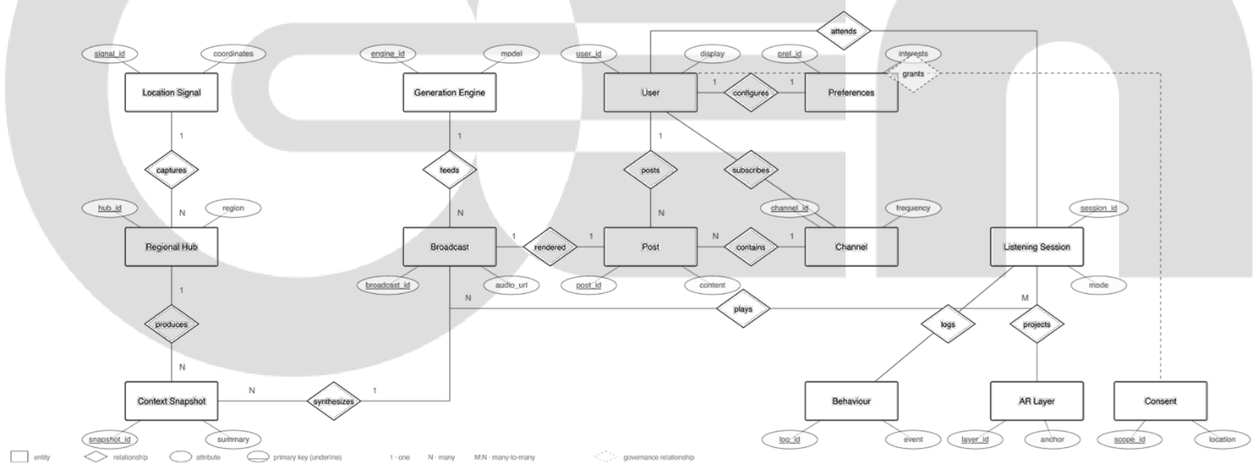
The AAR app shifts the layer of everyday life from visual to audio. It delivers a deeply personalized radio experience where AI learns how the user moves through the world and curates a living soundtrack around it. Not a playlist. Not a podcast. A radio that listens to you, so it knows what to play next.

- Communication starts from a basic algorithm and evolves autonomously via short encrypted identifiers and parsing-based request systems.
- Ambient, always-on delivery — the app is background audio infrastructure, not a foreground app.
- Smart-glasses integration provides direct audio output for wearable tech.

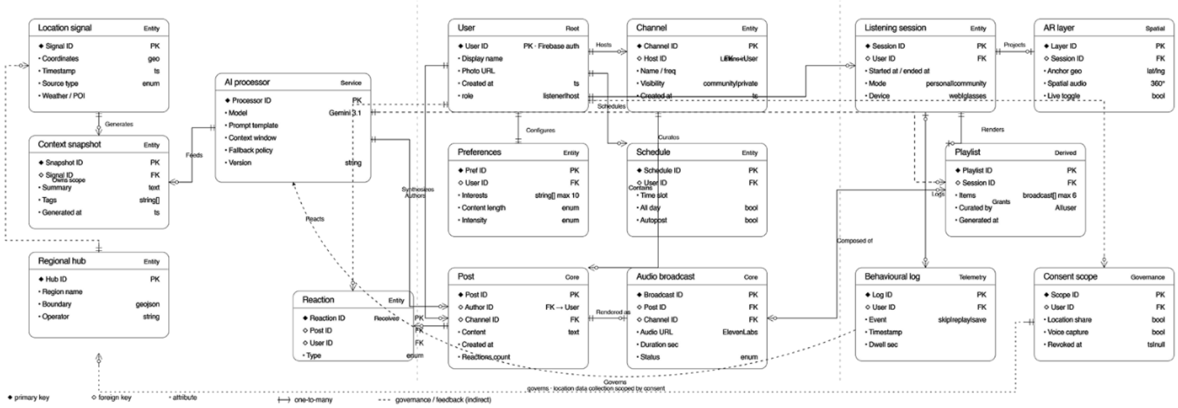



## Data Architecture

The live prototype runs on a fourteen-entity data model derived from the web implementation. Cross-boundary relationships connect the physical layer through broadcast synthesis to personalized delivery, governed end-to-end by ConsentScope.




Conceptual model showing ten core entities and their relationships across the three service components.




Technical model derived from the current web prototype.

## Public Pilot — GradEx 111

In the first public pilot at GradEx 111, the project broadcasts via YouTube with a live feedback loop implemented in real time. It serves both as a welcoming space for visitors to the OCAD Industrial Design community and as a multifaceted channel carrying the voice of that community.

- Visual radio streaming — on-site footage and community updates stream as a visual-radio format, not a podcast, maximizing immersion without replacing the audio channel.
- Live feedback loop — visitor reactions cycle back into the broadcast in real time.
- Validation scope — a single event, with results feeding the Phase II open-beta closed pilot (50–100 participants).

## Known Constraints

The project operates within three structural constraints, each a deliberate trade-off prioritizing validation over production readiness.

- Web-based implementation · Native AR is not yet implemented; a web application stands in as a foundational prototype.
- Gemini API dependency · Core AI functions rely on external services, limiting full control over processing and data handling.
- Pilot scale · Validation through a single event (GradEx 111) restricts the generalizability of results.

These constraints are scoping decisions, not failures. Each is scheduled to be addressed as the project transitions from academic validation to commercial deployment.

# 10 Business Model

OnGen sustains itself by converting real-time location context into advertising assets and by licensing community infrastructure to enterprise partners. The business model is designed for the era of AR glasses and wearables, curating raw urban data into spatial audio that fosters community identity and connection.

## Business Model Canvas

| Key Partners | Key Activities | Value Propositions |
|---|---|---|
| Location Data Providers — APIs for hyperlocal context (POI, events, weather).<br>Hardware Mfgs — Meta Ray-Ban, Apple Vision Pro, mobile OEMs for AAR integration.<br>Community / Cultural Institutions — OCAD University (GradEx 111) for user testing and spatial exhibitions. | Spatial Audio Rendering — creating 360° immersive radio environments.<br>Content Curation — automating the transformation of raw data into broadcast-ready audio.<br>Contextual Translation — processing visual / GPS data into real-time audio narratives. | For Individuals — a personalized radio station narrating the world around the user in real time.<br>For Communities — a collective voice that bridges individual insights into shared radio experiences.<br>AAR Experience — audio AR that bridges the physical and digital. |
| **Customer Relationships** | **Customer Segments** | **Key Resources** |
| Co-Creation — users evolve the service as they contribute personal journals and data.<br>Follower-Broadcaster — a social dynamic where users tune in to each other's spatial vibes. | Urban Explorers — people who want a narrated, social layer to their city experience.<br>Local Communities — campus or coworking groups looking for shared spatial audio.<br>AAR Early Adopters — users with smart glasses or those seeking heads-up interaction. | AAR Media Engine — proprietary logic for processing reality into audio.<br>Spatial Interaction Design — expertise in non-visual UI / UX.<br>Brand Identity — radio-station aesthetic and spatial interaction design. |
| **Channels** | **Cost Structure** | **Revenue Streams** |
| Web Application — primary touchpoint (ongenradio.web.app).<br>Smart Glasses Integration — direct audio output for wearable tech.<br>Social Sharing — AI-curated audio shared across platforms. | Computing & Inference — server expenses for real-time audio generation.<br>R&D — integration with AR devices and design engineering.<br>Operational — marketing, branding, and exhibition maintenance. | Premium Subscription — ad-free, unlimited credits, private channels.<br>B2B Community Licensing — automated private Community Radio networks.<br>Location-Based Audio Channels — hyperlocal ad revenue from contextually relevant audio. |

## Cost Structure — Operational Detail

- Infrastructure & Hosting · managed hosting fees for high-performance GPU servers running local LLMs and audio-to-audio workflows; storage for high-fidelity raw journals and generated podcasts.
- R&D and Tech Stack · local LLM optimization for low-latency natural voice synthesis; usage-based AI audio API licenses through partnerships.
- Content Moderation · automated and manual systems to keep the social-radio layer free of hate speech and inappropriate content; cybersecurity consulting to maintain anonymity and audio-persona encryption.

## Revenue Structure

A revenue model centered on individual memberships and corporate infrastructure supply, converting real-time location context into advertising assets.

- Premium Subscription · ad-free experience, unlimited credits for deeper interactions, ability to host private channels.
- B2B Community Licensing · licensing automated private Community Radio networks to organizations, plus data infrastructure for AI enterprises.
- Location-Based Audio Channels · hyperlocal ad revenue from contextually relevant, sponsored audio narratives based on real-time location and interest.

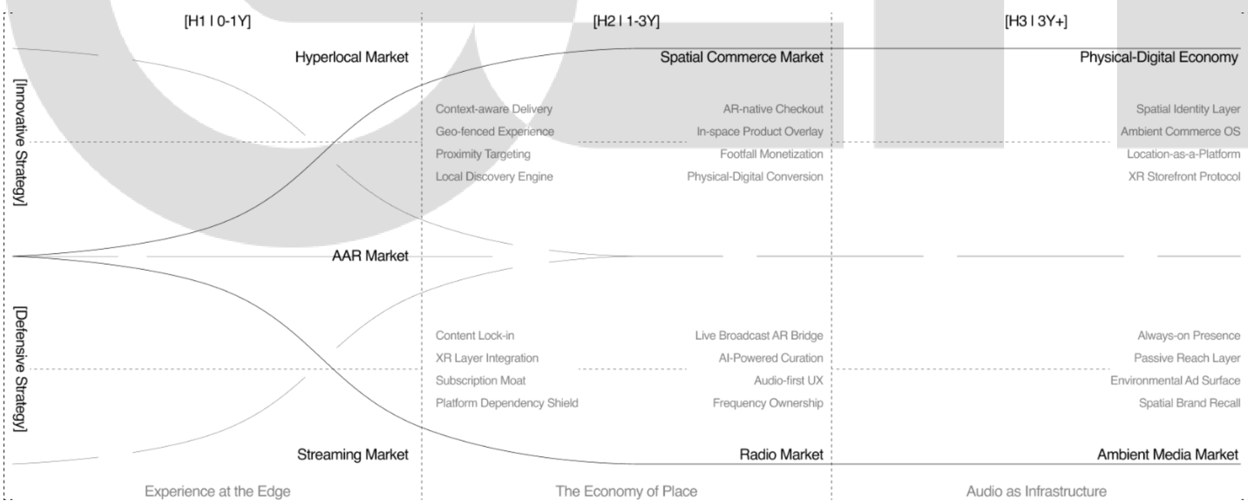
## Market Entry Roadmap

A three-horizon strategy pre-occupies the software and infrastructure-centric Audio-first AR market before AR glasses become mainstream, establishing a core market layer in preparation for the 2028 hardware explosion.

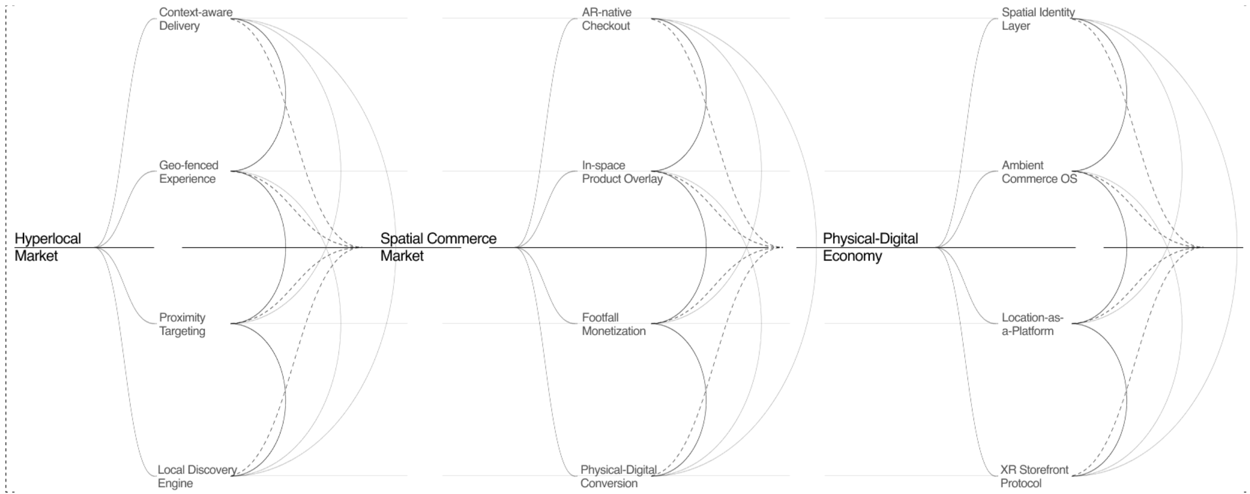
H1 · 0–1Y · Hyperlocal Market · Every physical location becomes a living broadcast channel delivering AI-generated audio experiences anchored to place, context, and the people within it.

H2 · 1–3Y · Spatial Commerce Market · Spaces gain economic identity as programmable media surfaces where presence drives commerce, attention, and community.

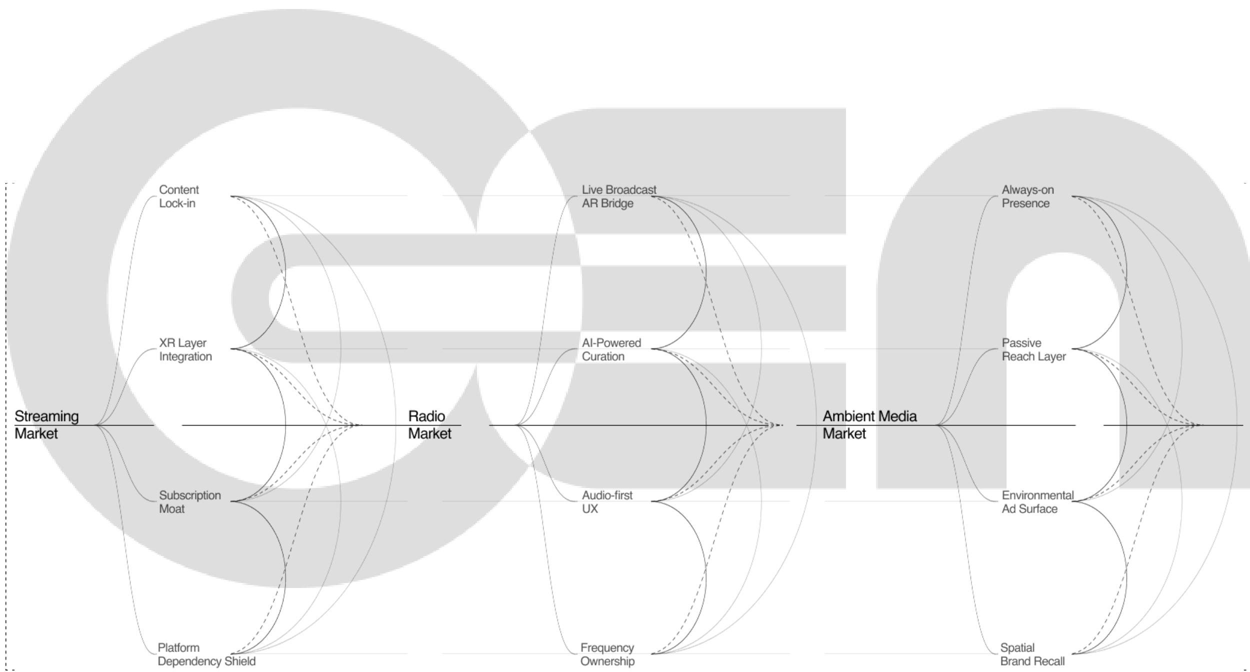
H3 · 3Y+ · Physical-Digital Economy · The boundary between physical and digital dissolves. Spatial radio becomes foundational infrastructure — as universal and invisible as connectivity itself.




Market Entry Roadmap Overview




Market Strategy I – Innovative strategic (offensive expansion) scenario



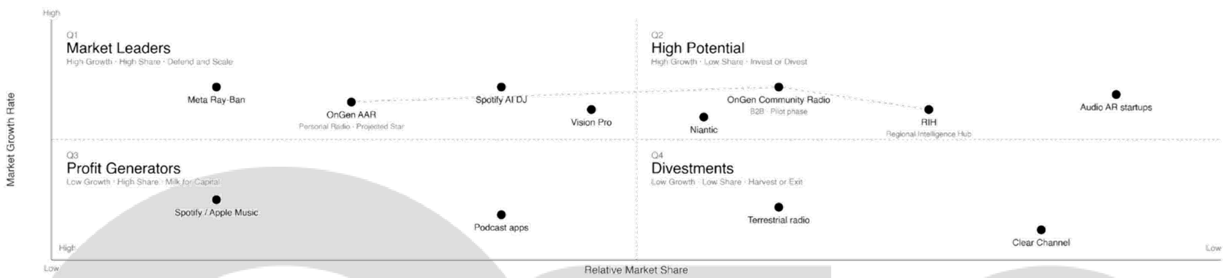

Market Strategy II – Defensive Strategic (Conservative expansion) scenario

## Strategic Analysis

OnGen's portfolio is stress-tested against four strategic frameworks to validate structural completeness and identify defensible assets.

### BCG Growth-Share Matrix

All three OnGen service components currently enter the market as Question Marks — high-growth category, no share yet. This is a capital-intensive phase that requires deliberate investment, with the AAR App targeted to transition into Star position as audio-AR adoption scales.



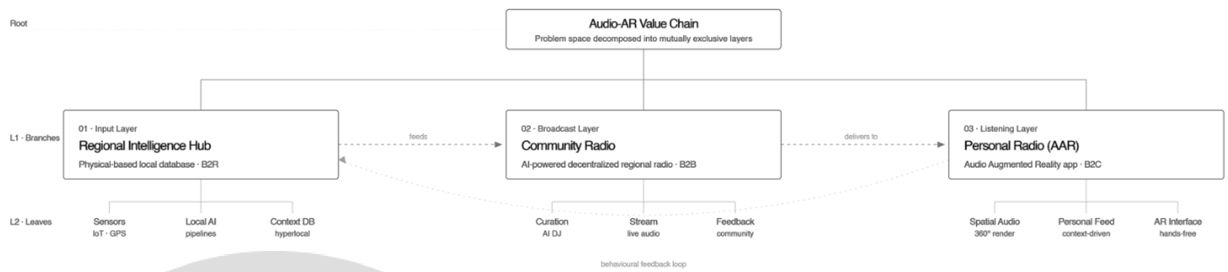

### VRIO Analysis

Of OnGen's key resources, two meet all four criteria of sustained competitive advantage: the AAR Media Engine and Community Partnerships. These warrant protection and compounding investment. The Gemini API dependency is the loudest red flag — valuable but exposed. Defending OnGen's moat means moving from rented intelligence toward owned inference over the next strategic cycle.

| Resource / Capability | Description | Valuable | Rare | Inimitable | Organized | Strategic Implication |
|---|---|---|---|---|---|---|
| AAR Media Engine | Audio-first AR translation of real-time context | Yes | Yes | Yes | Partial | Sustained Competitive Advantage. Priority asset, organize to capture |
| Spatial Interaction Design | Non-visual UI/UX expertise for ambient media | Yes | Yes | Partial | Yes | Temporary Competitive Advantage. Scarce talent pool, defend while it lasts |
| Brand Identity | Radio-rooted aesthetic, retro-futurist voice | Yes | Partial | Partial | Yes | Temporary Competitive Advantage. Differentiating today, copyable long-term |
| Community Partnerships | OCAD, cultural institutions, GradEx 111 pilot | Yes | Yes | Yes | Partial | Sustained Competitive Advantage. Relational moat, formalize governance |
| Location Data Pipeline | Hyperlocal POI, weather, event aggregation | Yes | No | No | Yes | Competitive Parity. Commodity layer, partner don't build |
| Gemini API Dependency | Core AI inference via external service | Yes | No | No | No | Competitive Disadvantage. Risk vector, plan migration path |

MECE Decomposition

The three service components are Mutually Exclusive and Collectively Exhaustive. Regional Intelligence Hub processes data, Community Radio broadcasts it, AAR App delivers it to the listener — no task performed by one is replicated by another. Together they cover the full chain of audio-AR delivery: physical context, processing, broadcast, listener experience, feedback. The behavioural feedback loop from AAR back into the Hub closes the cycle, meaning the portfolio is self-contained and structurally whole.




## KPIs — Internalization and Backgrounding

Three performance indicators track whether OnGen has successfully moved from a one-time experience to a weekly routine and, ultimately, to an invisible background layer.

| | | |
|---|---|---|
| 30% | Habit Formation Rate<br>7-Day Return Rate | Beyond a one-time experience, the service becomes a weekly routine — stronger retention than raw subscriber numbers. |
| ↑ | Audio-to-Session Ratio<br>Background Listen Duration | Securing a real share of the user's background. Unlike screen-based rivals, OnGen achieves invisible immersion by occupying time. |
| 3+ | Interaction-to-Listen Depth<br>Avg. Interactions per Session | Active response to audio content ensures high data density — a core asset for precision in future recommendations and ads. |

## Stakeholder Accountability

Because OnGen converts user-generated text into AI-driven live audio broadcast, the accountability chain is non-linear. A single community post passes through multiple stakeholders before reaching the listener: the user writes it, the AI reinterprets it, third-party services vocalize and deliver it, and the platform operator governs the entire pipeline. The table below maps where each stakeholder's responsibility begins and ends — and where the boundaries blur.

| Stakeholder | Accountability Area | Risk Zone |
|---|---|---|
| Platform Operator | Service operation, AI policy, content moderation, infrastructure. | AI DJ output quality, policy enforcement failure. |
| AI DJ / Curation Engine | Script generation, tone conversion, recommendation accuracy. | Distortion of user content, inappropriate broadcast. |
| Community Contributors | Content creation, community engagement. | Harmful or misleading content entering the broadcast pipeline. |
| Third-party Services (TTS, Firebase) | Service availability, data security, SLA compliance. | Downtime, data breach, latency in real-time broadcast. |

# Conclusion

OnGen Radio is a proposition that the next layer of human–AI experience will not arrive through another screen but through the audio channel that already sits in the user's ears, in the background, hands-free and eyes-free. It repositions radio — long written off as legacy media — as foundational infrastructure for the ambient, generative, context-aware era that AR hardware will require.

The project is not a finished product. It is a validated system framework — three mutually exclusive service components, one closed feedback loop, five brand values, three perceptual-market positions, and a three-horizon roadmap — scoped deliberately to prioritize validation over production readiness. The web-based pilot, the Gemini API dependency, and the single-event scale are trade-offs that leave every subsequent decision traceable and every constraint addressable.

What began as a thesis statement — design is not a fixed result; design is the potential of possibilities through adaptation — ends as an architecture. The architecture listens, speaks, restrains itself, and leaves room for what it cannot yet know. The radio is reborn, quieter and more local than before, and the signal belongs to the people in the room.

— Michael Joongmin Park

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